



So about that Quantum Lattice Thing: Rebuttal to “Exact Coset Sampling for Quantum Lattice Algorithms”

Daniel Apon
Anduril Industries
dapon@anduril.com

October 28, 2025

Abstract

In this note we identify major issues with “Exact Coset Sampling for Quantum Lattice Algorithms” by Zhang. The issues include:

- in the first version, the proposed algorithm requires foreknowledge of the answer to compute the answer;
- in the latest version, the proposed algorithm displays basic misunderstandings of quantum mechanics.

We believe the existence of these issues refute the author’s claims.

Update: October 28, 2025. In response to [Zha25a], posted October 24, 2025,

1. We add a brief technical section at the end of this document, cf. Section 4.
2. We no longer plan to update this document further.
3. We strongly encourage the author to either:
 - (a) add to his various arXiv and GitHub pre-prints the following text, prominently displayed on the first page of each document:

Produced by humans. The author hereby confirms that the content in this pre-print writeup: (i) was produced by the author as his novel intellectual contribution, and (ii) was not produced by generative artificial intelligence (AI), with the possible exception of AI-proposed grammar improvements or minor integrated suggestions.
 - (b) or: in accordance with IACR [PKC26], ACM [ACM25], and IEEE [IEE25] policies, add a statement of Use of AI, prominently displayed, which records which LLM and version number were used, along with a full description of each query to the LLM that helped to generate the technical content of the pre-print.

1 Background

Chen’s preprint. On April 10, 2024 (the first day of the 5th NIST PQC Standardization Conference), Professor Yilei Chen posted a paper on IACR ePrint titled “Quantum Algorithms for Lattice Problems” [Che24]. The initial claim was a new quantum polynomial algorithm for solving Learning With Errors (LWE), which is the presumed hard computational problem underlying the majority of the Post-Quantum Cryptography (PQC) standards selected by the National Institute of Standards and Technology (NIST) as of July 5, 2022 [AAC⁺22, NIST24a, NIST24b]. Notably, this would include ML-KEM, ML-DSA, the upcoming FN-DSA standard, and all known fully homomorphic encryption (FHE) schemes.

This kicked off an intense period of attention where many parties around the world carefully analyzed the new claim. The result was a major bug in the approach found in Step 8.g, with credit to Thomas Vidick and Hongxun Wu, showing that the approach there does not (yet) work. This is properly well-documented on ePrint by [Che24].

Zhang’s preprint. Now a year and a half later, just preceding the 6th NIST PQC Standardization conference (which began on September 24, 2025), Yifan Zhang posted a paper on arXiv on September 17, 2025 titled “Exact Coset Sampling for Quantum Lattice Algorithms” [Zha25b]. The fresh claim here was a new, “simple, fully correct, and assumption-light replacement for the contested ‘domain-extension’ in Step 9 [by Chen];” that is, that there is a claimed fix or work-around to the barriers in Chen’s approach.

Here, we aim to provide a counter-analysis of this novel claim.

1.1 Contributions of this short note

In this short note, we raise a variety of technical objections to Zhang’s new work. In other words:

We do not believe the newly modified quantum attack against lattice cryptography is correct.

Much thought has gone into whether to publish some note of any kind.

In particular, we are of the general opinion that original, highly technical material should first go through a blind peer-review process, e.g. in a well-respected IACR conference. However, we have decided to write this, given the high stakes of the existence of a provably correct quantum polynomial time algorithm solving Learning With Errors problems, given that the new document has been disseminated online, and given that there appears no other viable forum for which to discuss the technical merit of the work.

Further, we want to point out our belief that developing new quantum algorithms from new techniques or ideas in order to solve hard computational problems is of the utmost importance. This note is not intended to dissuade further research into quantum algorithms.

As is typical in cryptography, just because one path (or paths) may not work out does not mean that some other new path cannot. ~~We intend to update this document in response to any new events.~~ Currently, we expect no new, significant, technical advances are coming in the immediate future.

1.2 Roadmap

In Section 2, we recite the technical situation and set up the discussion.

In Section 3, we provide two technical objections, in order of increasing severity.

2 Our Rebuttal: The Technical Set-Up

We dive into the detailed matter of the new quantum algorithm claim. We will provide two main objections. They are arranged in order of increasing severity.

The reader is asked to keep in mind that the initial arXiv draft by Zhang has gone through multiple revisions in a very short amount of time.¹ We will speak to the first version in the first objection, then to the final version in the latter.

We begin with a recitation of the essential elements of the proposed algorithms, first by Chen and then by Zhang.

2.1 Overview of the old (Chen) and new (Zhang) algorithms

2.1.1 Recitation of Chen's algorithm

Chen's proposed (and retracted) quantum algorithm for lattice problems [Che24] has the following structure:

In Step 1 (pages 22-23 of [Che24]), one begins (call it Step 1.1) by preparing a uniform superposition over a modular lattice L (given basis matrix $\mathbf{A}$ for L as input) as well as a complex Gaussian state representing, roughly, a Gaussian-weighted superposition over an n -dimensional ball centered at the origin. This is expressed as (Equation 16 [Che24]):

$$\sum_{\mathbf{v} \in L \cap \mathbb{Z}_{Dq}^n} |\mathbf{v}\rangle \otimes \sum_{\mathbf{y} \in \mathbb{Z}^n \cap (r \log n) \mathcal{B}_{\infty}^n} \rho_r(\mathbf{y}) \cdot e^{\frac{-\pi i \|\mathbf{y}\|^2}{s^2}} |\mathbf{y}\rangle.$$

Then (Step 1.2), one adds the first register to the second, yielding

$$\sum_{\mathbf{v} \in L \cap \mathbb{Z}_{Dq}^n} |\mathbf{v}\rangle \otimes \sum_{\mathbf{y} \in \mathbb{Z}^n \cap (r \log n) \mathcal{B}_{\infty}^n} \rho_r(\mathbf{y}) \cdot e^{\frac{-\pi i \|\mathbf{y}\|^2}{s^2}} |\mathbf{y} + \mathbf{v} \bmod Dq\rangle.$$

Then (Step 1.3), a measurement is performed on the second register $|\mathbf{y} + \mathbf{v} \bmod Dq\rangle$ yielding a classical string $\mathbf{y}' \in \mathbb{Z}_{Dq}^n$ and a residual single register $|\mathbf{v}\rangle$ whose amplitudes have been re-normalized.

Finally (Step 1.4), one computes in the first register the value $|\mathbf{v} - \mathbf{y}' \bmod Dq\rangle$.

The resulting state at the end of Step 1 is (properly) analyzed to be (close to)

$$|\psi'_1\rangle = \sum_{k \in \mathbb{Z}} \alpha_k |k\mathbf{x} - \mathbf{y} \bmod P\rangle, \tag{1}$$

¹Indeed, we wanted to intentionally wait until some draft had stabilized for a little while before writing this note.

for some amplitudes α_k that are not relevant to this note (and, in particular, are entirely unused and untouched by [Zha25b]).

The remainder of Steps 2 to 8.f (or 8.g) in [Che24] manipulate this superposition $|\psi'_1\rangle$ in order to produce a so-called “Karst wave” effect in the amplitudes α_k (noting that the index of superposition k here is equivalent by change of variables via QFT/PSF to another index variable j later on).²

After a long calculation and manipulation in [Che24] we arrive at the state (Equation 40 in [Che24] or Equation 1.1 in [Zha25b]):

$$|\phi_{8.f}\rangle = \sum_{j \in \mathbb{Z}} \alpha(j) |2D^2 j b_1^* | 2D^2 j \mathbf{b}_{[2..n]}^* + \mathbf{v}_{[2..n]}^* \bmod M_2\rangle.$$

Simplifying notation for our purposes, we can treat the above constructed superposition as

$$|\phi\rangle = \sum_{j \in \mathbb{Z}} \alpha(j) |j \mathbf{b}^* + \mathbf{v}^* \bmod M_2\rangle, \quad (2)$$

noting – by simple inspection – the similarity in form between Equations (1) and (2). That is, so long as differences in amplitudes and the low-level details of the format of vectors $\mathbf{x}, \mathbf{y}$ vs $\mathbf{b}^*, \mathbf{v}^*$ are ignored (as these are not used by the ideas in [Zha25b]), they are linear equations in the index of superpositions (j and k , respectively) over unknown vectors $\mathbf{x}, \mathbf{y}$ vs $\mathbf{b}^*, \mathbf{v}^*$.

As mentioned before, the basic idea of manipulating the amplitudes from α_k to $\alpha(j)$ in Steps 2 to 8 in [Che24] was to create a particular periodic relationship – a “Karst wave” – that would allow extraction of important information regarding the unknown $\mathbf{b}^*$ (or, respectively, the unknown $\mathbf{x}$ in the first place). This approach has been shown not to work so far.

2.1.2 Recitation of Zhang’s proposed patch

The general idea of [Zha25b] is to produce a “pair-shift difference.” We quote the “idea in one line” description:

Make a second copy of the coordinate registers, coherently shift it by a uniform $T \in \mathbb{Z}_P$ along the $\mathbf{b}^*$ direction, and subtract. The subtraction cancels the unknown offsets $\mathbf{v}^*$ and leaves a clean difference register $-2D^2 T \mathbf{b}^* \bmod M_2$. [...]

In other words, given the term, or the ability to produce the term, looking like (and suppressing unnecessary constants/etc.):

$$\sum_{j \in \mathbb{Z}} \alpha(j) |j \mathbf{b}^* + \mathbf{v}^*\rangle,$$

find a way to compute another term

$$\sum_{j \in \mathbb{Z}} \alpha(j) |(j+1) \mathbf{b}^* + \mathbf{v}^*\rangle,$$

whose subtraction yields

$$\sum_{j \in \mathbb{Z}} \alpha(j) |\mathbf{b}^*\rangle. \quad (3)$$

²A key contribution of this note is to point out to the broader cryptographic community that the analysis of the current situation can be focused only on Step 1 of Chen’s algorithm, and may ignore everything from Step 2 onward.

From this point, the new algorithm in [Zha25b] proceeds to clean up various matter and aim to conclude with the original goal of [Che24] (extracting information that breaks LWE).

3 Our Rebuttal: Two main objections

As hinted at above, our rebuttal (and objections) center only around the very first steps of the algorithmic process detailed in [Zha25b]. We strongly believe that once one can compute a superposition corresponding to Equation (3) that the main work of the subroutine is complete, and the rest can be accomplished in any manner of different ways – for example, as given in the preprint.

In the following, we will now delineate between which versions of “Exact Coset Sampling for Quantum Lattice Algorithms” to which we’re referring. The original version was posted on arXiv on September 15, 2025, linking to the project page

<https://github.com/yifanzhang-pro/quantum-lattice>.

After some early personal communication with the author, the paper has been updated many times (e.g. going from 7 pages to 18 pages) in a couple of weeks.

We will refer to the initial version [Zha25b] when discussing Issue 1, and then to the most recent version we have found [Zha25c] when discussing Issue 2.

3.1 Issue 1: Assuming the answer to compute the answer, in the first version

Issue 1 is fairly straightforward.

In this initial version of the preprint, Step 9^{†.1} (copy) begins by copying the following, using a CNOT:

$$\sum_j \alpha(j) |\mathbf{X}(j)\rangle |0\rangle \rightarrow \sum_j \alpha(j) |\mathbf{X}(j)\rangle |\mathbf{X}(j)\rangle,$$

where – just as a matter of notation – we write $\mathbf{X}(j) := \left(2D^2 j b_1^* \parallel 2D^2 j \mathbf{b}_{[2..n]}^* + \mathbf{v}_{[2..n]}^* \right)$.

Then critically, Step 9^{†.2} (coherent P -shift along $\mathbf{b}^*$) asks for the following operation:

- Add $2D^2 T \mathbf{b}^*$ to $\mathbf{Y}$ (the second register, a copy of the first) coordinate-wise:

$$|\mathbf{X}(j)\rangle |\mathbf{Y}\rangle |T\rangle \rightarrow |\mathbf{X}(j)\rangle |\mathbf{Y} + 2D^2 T \mathbf{b}^*\rangle |T\rangle$$

The issue, of course, is that computing the term $2D^2 T \mathbf{b}^*$ (or even knowing the term!) in order to add it into the second register via (modular) addition gates at this step assumes knowledge of the answer for which you’re seeking.

It should be clear that knowing $\mathbf{b}^*$ is sufficient to “win.” To verify this, observe that the goal of Step 9 is to generate a random vector $\mathbf{u} \in \mathbb{Z}_{M_2}^n$ that satisfies the modular linear relation

$$\langle \mathbf{b}^*, \mathbf{u} \rangle \equiv 0 \pmod{P}.$$

Given knowledge of $\mathbf{b}^*$, this is simple.³ But “simply adding” a term dependent on $\mathbf{b}^*$ to some register, requires knowing $\mathbf{b}^*$ in the first place.

³Use Gaussian elimination.

Looking at Appendix B in [Zha25b], we see that – apparently – the term $2D^2\mathbf{b}^*$ should already be precomputed. But it is not explained how to do so. (Nor is it well-explained why one hasn't already short-cut the entire algorithm of Chen, and just worked from Step 1, since you've “precomputed” an explicit function of $\mathbf{b}^*$ – which is the goal in the first place.)

Thus presumably, there were the revisions to the online version(s) to explain this. We continue.

3.2 Issue 2: From its beginning intuition, the proposed algorithm displays basic misunderstandings of quantum mechanics

Here, we provide our main objection.

The conceptual/intuitive essence of the pair-shift difference idea motivating [Zha25b, Zha25c] comes down to the existence of a so-called coordinate evaluator U_{coords} , some reversible arithmetic block that writes the coordinate registers appearing in the superposition

$$|\phi_{8.f}\rangle = \sum_{j \in \mathbb{Z}} \alpha(j) \left| 2D^2jb_1^* \parallel 2D^2j\mathbf{b}^* + [2..n] + \mathbf{v}_{[2..n]}^* \bmod M_2 \right\rangle,$$

on basis input j as:

$$U_{coords} : |j\rangle|\mathbf{0}\rangle \rightarrow |j\rangle|\mathbf{X}(j)\rangle,$$

where U_{coords} is only called on basis inputs (e.g. $j = 0, 1$) to gather the initial critical data.

Observation 1. First, we make a simple statement. By definition,

$$U_{coords} : |j\rangle|\mathbf{0}\rangle \rightarrow |j\rangle|\mathbf{X}(j)\rangle,$$

is an entirely classical circuit. (It is a reversible arithmetic block!) Yet, the map $|\mathbf{0}\rangle \rightarrow |\mathbf{X}(j)\rangle$ is clearly intended to replicate the work of Chen [Che24] between its initial state preparation and the superposition at Step (8.f), which presumably we want to treat as a quantum algorithmic pipeline.⁴

We elaborate on this key point further below, by explicit technical demonstration.

Observation 2. We claim that the computation from Step 1 to Step (8.f) in Chen's approach is inherently, arithmetically non-reversible (and thus, “quantum”) – from its earliest step. Therefore, there is no evidence that an efficient U_{coords} even exists!

As detailed in Section 2.1.1, one begins by (validly!) producing a superposition

$$\sum_{\mathbf{v} \in L \cap \mathbb{Z}_{Dq}^n} |\mathbf{v}\rangle \otimes \sum_{\mathbf{y} \in \mathbb{Z}^n \cap (r \log n) \mathcal{B}_\infty^n} \rho_r(\mathbf{y}) \cdot e^{\frac{-\pi i \|\mathbf{y}\|^2}{s^2}} |\mathbf{y} + \mathbf{v} \bmod Dq\rangle. \quad (4)$$

then **measures** it.

The intuition behind this beginning step is to first create a superposition over a modular lattice (i.e. $|\mathbf{v}\rangle$) and a superposition over a short infinity-ball (i.e. $|\mathbf{y}\rangle$), then sum them to produce a superposition over an LWE-like expression $|\mathbf{y} + \mathbf{v} \bmod Dq\rangle$.

⁴Or better yet, just show the classical reversible circuit that efficiently computes this!

Prior to the measurement of the second register in Equation (4), both $|\mathbf{v}\rangle$ and $|\mathbf{y} + \mathbf{v} \bmod Dq\rangle$ are coherent quantum states. By definition [NC00], this means that they are individually unit-norm vectors each in a high-dimensional complex vector space, and the joint quantum state lives in a tensor product of two complex vector spaces – call this space $V \otimes W$ so that (dropping unnecessary notation):

$$|\mathbf{v}\rangle \otimes |\mathbf{y} + \mathbf{v}\rangle \in V \otimes W.$$

Upon measuring the second register of this tensor-state, the complex vector $|\mathbf{y} + \mathbf{v}\rangle$ (that contains quantum information) is projected from the ambient vector space W to some subspace of W (that only contains classical information – in this case $\mathbf{y}'$ – which is output classically to the measurer). Then as an analytic result, the amplitudes of the factor-vector $|\mathbf{v}\rangle$ are necessarily re-normalized (so that probabilities sum to 1) and so $|\mathbf{v}\rangle$ lives in the factor-vector-space V alone.

Measurement is a projection operator [NC00]. Projection is not a reversible operation! This fools the obvious existence of a clean, efficiently computable, reversible U_{coords} unitary in any black box sense, which destroys⁵ the newly proposed quantum lattice algorithm.

4 Comments on [Zha25a] (October 28, 2025)

The revised response makes a single claim that relates to the rebuttal of this document: that deferred measurement resolves the above reversibility issue.

It raises another killer problem: In this new circuit, one must now make two measurements that are strongly correlated. In other words, measuring the two (one, copied) registers must result in the same

$$|k\mathbf{x} - \mathbf{y} \bmod P\rangle$$

in Equation 1, rather than a first $(\mathbf{x}, \mathbf{y})$ and a second, distinct $(\mathbf{x}', \mathbf{y}')$.

This event (i.e., that $\mathbf{x} = \mathbf{x}'$ and/or $\mathbf{y} = \mathbf{y}'$) occurs with only negligible probability.

Acknowledgments

We would like to thank those with which we've had many useful discussion in the course of preparing this note, especially those at the 6th NIST PQC Standardization Conference's "hallway track." We especially thank Brendan Zember (Univ of Maryland, College Park) for first pointing out the new paper published on arXiv, and for the many, long, fruitful technical discussions with him on the topic area.

References

- [AAC⁺22] G. Alagic, D. Apon, D. Cooper, Q. Dang, T. Dang, J. Kelsey, J. Lichtinger, Y. Liu, C. Miller, D. Moody, R. Peralta, R. Perlner, A. Robinson, and D. Smith-Tone. Status Report on the Third Round of the NIST Post-Quantum Cryptography Standardization Process. NIST Interagency/Internal Report (NISTIR), 2022. Available online at: <https://nvlpubs.nist.gov/nistpubs/ir/2022/NIST.IR.8413-upd1.pdf>. Last accessed October 13, 2025.

⁵This is the correct spelling.

- [ACM25] ACM. ACM Policy on Authorship, 2025. Available online at: <https://www.acm.org/publications/policies/frequently-asked-questions>. Last accessed October 28, 2025.
- [Che24] Yilei Chen. Quantum Algorithms for Lattice Problems. Cryptology ePrint Archive, Paper 2024/555, 2024. Available online at: <https://eprint.iacr.org/2024/555>. Last accessed October 13, 2025.
- [IEE25] IEEE. IEEE Publication Services and Products Board Operations Manual 2025, 2025. Available online at: <https://pspb.ieee.org/images/files/PSPB/opsmanual1.pdf>. PDF Page 106. Last accessed October 28, 2025.
- [NC00] Michael A. Nielsen and Isaac L. Chuang. Quantum Computation and Quantum Information. Cambridge University Press, 2000.
- [NIST24a] National Institute of Standards and Technology. Module-Lattice-Based Digital Signature Standard. Federal Information Processing Standards (Federal Register), 2024. Available online at: <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.204.pdf>. Last accessed October 13, 2025.
- [NIST24b] National Institute of Standards and Technology. Module-Lattice-Based Key-Encapsulation Mechanism Standard. Federal Information Processing Standards (Federal Register), 2024. Available online at: <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.203.pdf>. Last accessed October 13, 2025.
- [PKC26] PKC. IACR AI Tool Policy, 2026. Available online at: <https://pkc.iacr.org/2026/aipolicy.php>. Last accessed October 28, 2025.
- [Zha25a] Yifan Zhang. A Note on Apon (2025)’s Comment on Quantum Lattice Algorithms, 2025. Available online at: https://github.com/yifanzhang-pro/quantum-lattice/blob/master/docs/Note_on_Daniel_Apon.pdf. Last accessed October 28, 2025.
- [Zha25b] Yifan Zhang. Exact Coset Sampling for Quantum Lattice Algorithms (First publication; September 15, 2025), 2025. Available online at: <https://arxiv.org/abs/2509.12341v1>. Last accessed October 13, 2025.
- [Zha25c] Yifan Zhang. Exact Coset Sampling for Quantum Lattice Algorithms (Latest version; October 1, 2025), 2025. Available online at: <https://arxiv.org/abs/2509.12341v3>. Last accessed October 13, 2025.